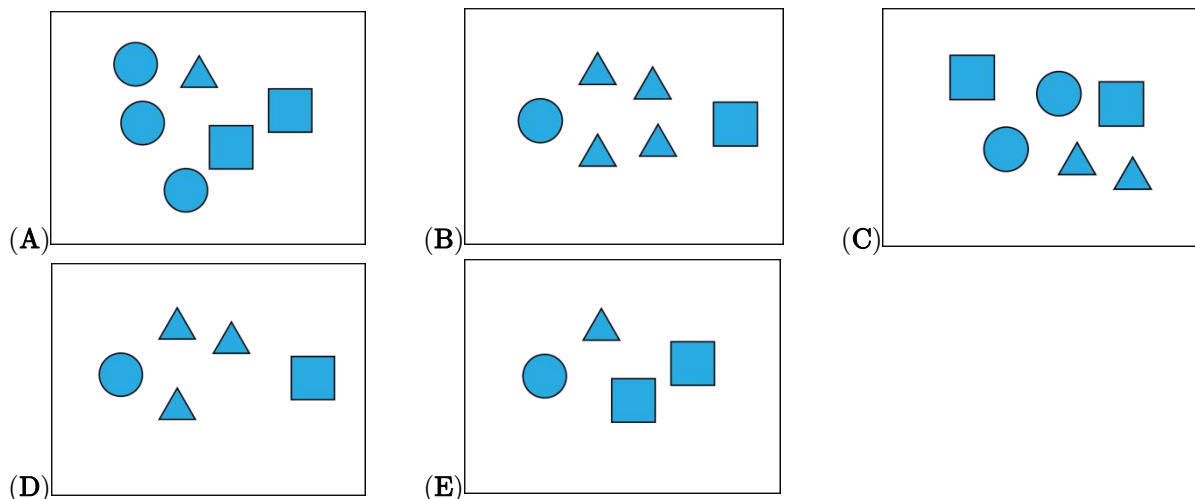


3 points

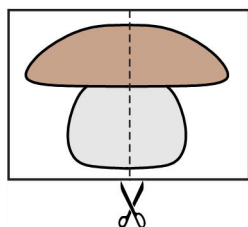
1. Which box contains the most triangles?

xià miàn nǎ ge hé zǐ lǐ de sān jiǎo xíng zuì duō
下 面 哪 个 盒 子 里 的 三 角 形 最 多?



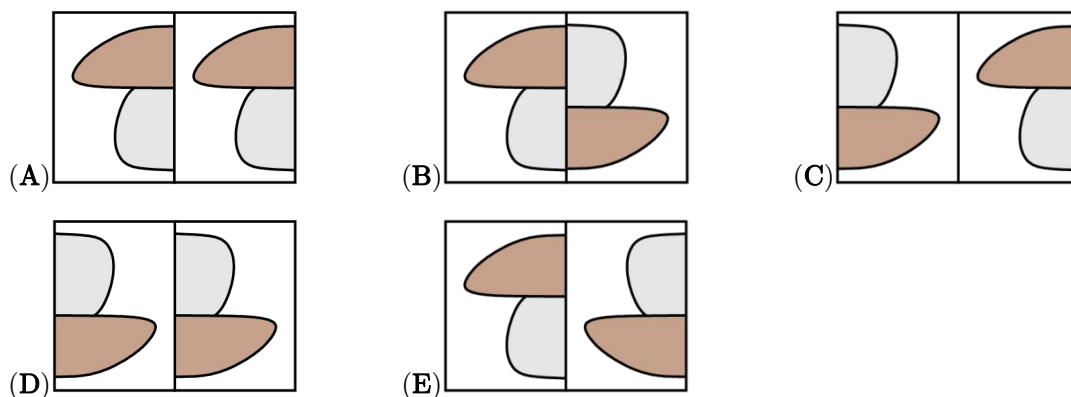
2. Arek cuts this picture in half and puts the two pieces together.

ā léi jiāng xià miàn zhè zhāng zhào piàn jiǎn chéng liǎng bàn rán hòu chóng xīn pīn jiē zài yì qǐ
阿 雷 将 下 面 这 张 照 片 剪 成 两 半, 然 后 重 新 拼 接 在 一 起。



Which option shows the two pieces of Arek's picture?

xià miàn nǎ ge xuǎn xiàng shì ā léi pīn jiē chū lái de xíng zhuàng
下 面 哪 个 选 项 是 阿 雷 拼 接 出 来 的 形 状 ?

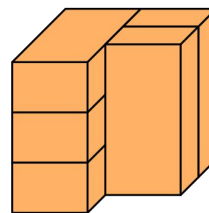


3. The picture shows 5 identical bricks.

How many bricks are touching exactly 3 other bricks?

tú zhōng yǒu 5 kuài xiāng tóng de zhuān
图中有5块相同的砖。

qí zhōng yǒu jǐ kuài zhuān qià hǎo āi zhe líng wài 3 kuài zhuān
其中有几块砖恰好挨着另外3块砖？

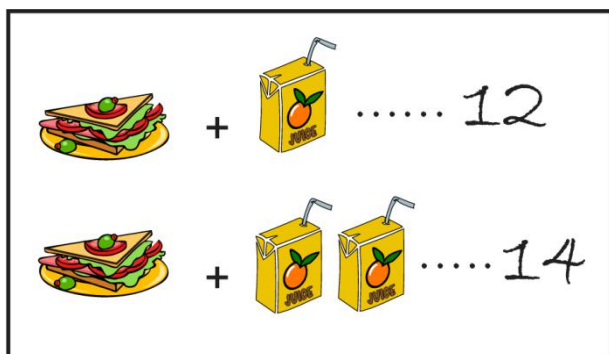


- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

4. One sandwich and one juice together cost 12 euro. One sandwich and two juices together cost 14 euro.

yī fèn sān míng zhì hé yī bēi guǒ zhī jiā qǐ lái gòng 12 yuán
一份三明治和一杯果汁加起来共12元，一份三明治和两杯果汁

jiā qǐ lái gòng 14 yuán
加起来共14元。



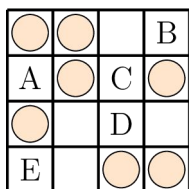
How much does one juice cost?

nà me yī bēi guǒ zhī duō shǎo qián
那么一杯果汁多少钱？

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

5. There has to be 2 coins in each row and each column.

yào shǐ měi háng hé měi liè dōu bì xū yǒu 2 méi yìng bì
要使每行和每列都必须有2枚硬币。



Where do you need to put the final coin?

nà me zuì hòu yī méi yìng bì yīng gāi fàng zài nǎ ge wèi zhì
那么最后一枚硬币应该放在哪个位置？

- (A) A (B) B (C) C (D) D (E) E

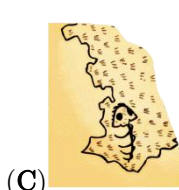
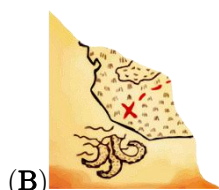
6. A monkey has torn a piece from Captain Jack's map.

yī zhī hóu zi bǎ jié kè chuán zhǎng de dì tú sī diào le yī kuài
一只猴子把杰克船长的地图撕掉了一块。



Which is the missing piece?

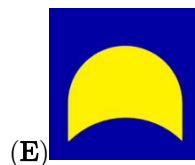
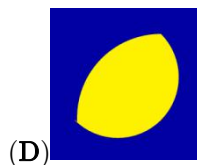
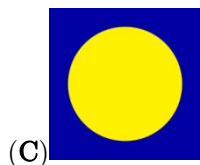
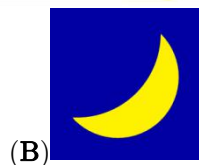
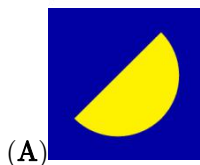
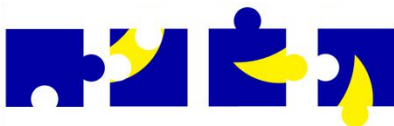
xià miàn nǎ kuài shì dì tú quē shī de bù fēn
下面哪块是地图缺失的部分?



7. Peter puts the 4 puzzle pieces shown together to make a square. Which picture can he make?

bǐ dé jiāng xià miàn piàn pīn tú pīn zài yì qǐ zǔ chéng yí gè zhèng fāng xíng qǐng wèn tā néng
彼得将下面4片拼图拼在一起组成一个正方形。请问他能

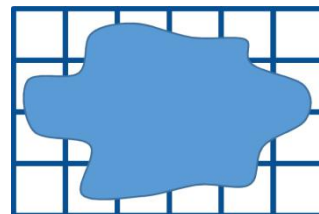
pīn chū xuǎn xiàng zhōng de nǎ fú tú
拼出选项中的哪幅图?



8. Some ink spilled on a piece of squared paper, as shown in the picture.
How many of the squares have ink on them?

rú tú suǒ shì mò shuǐ sǎ zài le yī zhāng fāng gé zhǐ shàng
如图所示,墨水洒在了一张方格纸上。

qǐng wèn yǒu duō shǎo gè xiǎo fāng gé zhān shàng le mò shuǐ
请问有多少个小方格沾上了墨水?



(A) 16

(B) 17

(C) 18

(D) 19

(E) 20

4 points

9. Kanga wrote down a number and then covered each digit with a shape.

Different digits were covered by different shapes, and the same digits were covered by the same shape.



Which number could be written under these shapes?

xiǎo dài shǔ xiě xià yī gè shù rán hòu yòng gè zhǒng xíng zhuàng gài zhù měi gè shù wèi shàng de
小袋鼠写下一个数, 然后用各种形状盖住每个数位上的

shù zì bù tóng xíng zhuàng xià fāng de shù zì bù tóng xiāng tóng xíng zhuàng xià fāng de shù zì
数字。不同形状下方的数字不同, 相同形状下方的数字

xiāng tóng
相同。

qǐng wèn zhè xiē xíng zhuàng xià fāng kě néng xiě yǒu nǎ ge shù
请问这些形状下方可能写有哪个数?

- (A) 34426 (B) 34526 (C) 34423 (D) 34424 (E) 32446

10. One animal sleeps in each of the baskets. The koala and the fox are sleeping in baskets with the same pattern and shape.

měi gè lán zi lǐ dōu yǒu yī zhī dòng wù zài shuì jiào kǎo lā hé hú lí shuì zài tú àn hé xíng
每个篮子里都有一只动物在睡觉。考拉和狐狸睡在图案和形

zhuàng dōu xiāng tóng de lán zi lǐ
状都相同的篮子里。



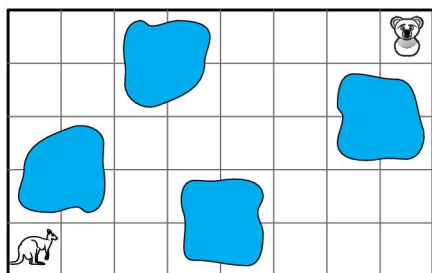
The kangaroo and the ostrich have the same pattern on their baskets. Which basket is the puppy sleeping in?

dài shǔ hé tuó niǎo shuì zài tú àn xiāng tóng de lán zi lǐ nà me xiǎo gǒu shuì zài nǎ ge lán zi
袋鼠和鸵鸟睡在图案相同的篮子里。那么小狗睡在哪个篮子
lǐ
里?

- (A) basket 1 1 hào lán zi (B) basket 2 2 hào lán zi (C) basket 3 3 hào lán zi
(D) basket 4 4 hào lán zi (E) basket 5 5 hào lán zi

11. Kanga wants to reach the koala without going through any of the coloured squares.

dài shǔ xiǎng zhǎo kǎo lā wán, dàn bù xiǎng jīng guò rèn hé yī gè cǎi sè fāng kuài.
袋鼠想找考拉玩，但不想经过任何一个彩色方块。



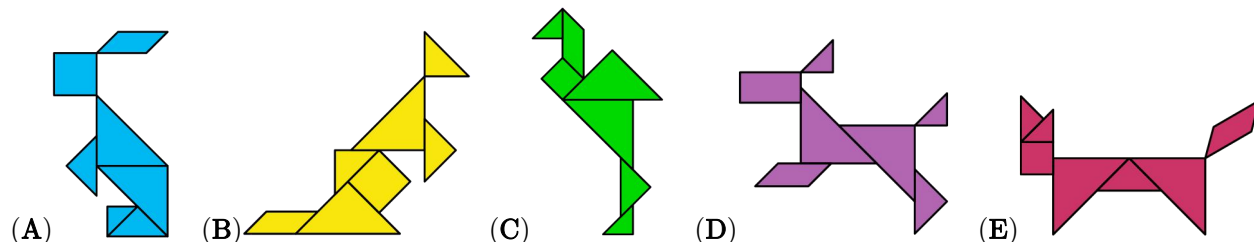
Which route could she take?

tā kě yǐ zǒu xià miàn nǎ tiáo lù xiàn?
她可以走下面哪条路线?

- (A) → → ↑ ↑ → → ↑ ↑ → →
- (B) → → ↑ ↑ → → → → → ↑ ↑
- (C) → → ↑ ↑ ↑ ↑ → → → → →
- (D) → → ↑ ↑ → → ↑ ↑ ← ← ←
- (E) → → ↑ ↑ ↑ → → ↑ → → →

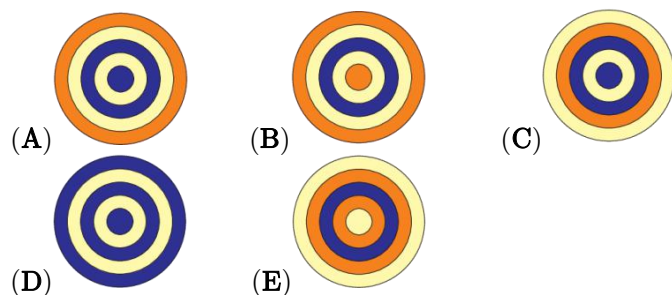
12. In one of the pictures below, a shape is used that cannot be seen in the others. In which picture is it?

xià miàn nǎ zhāng tú piàn zhōng, shǐ yòng le qí tā tú piàn zhōng méi yǒu yòng dào de xíng zhuàng?
下面哪张图片中，使用了其他图片中没有用到的形状?



13. Which option shows the view from above this stack of discs?

cóng shàng fāng kàn zhè dié yuán pán kě yǐ dé dào xià miàn nǎ fú tú?
从上方看这叠圆盘可以得到下面哪幅图?



14. Which of the following pictures will we see when we use the stamp shown?

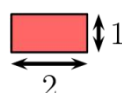
使用右图中的印章时,可以得到下面哪种印记?



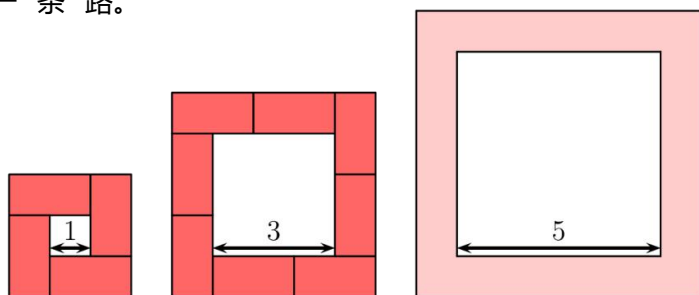
- (A) (B) (C) (D) (E)

15. Katrin builds a path around each square using tiles like the one shown.

小卡用如图所示的地砖
一条路。



在每个正方形周围都铺了

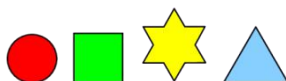


How many tiles does she use around a square with side 5?

请问她在边长为5的正方形周围铺路用了多少块地
砖?

- (A) 10 (B) 11 (C) 12 (D) 14 (E) 16

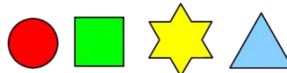
16. Ann has 4 stickers as shown.



She sticks down the star after she sticks down the square. She sticks down the star before she sticks down the triangle.

Which picture could she end up with?

安安有4张如图所示的贴纸。



她在贴完正方形之后贴了星形,在贴三角形之前先贴了星形。

她最终可能会得到下面哪张图片?

- (A) (B) (C) (D) (E)

5 points

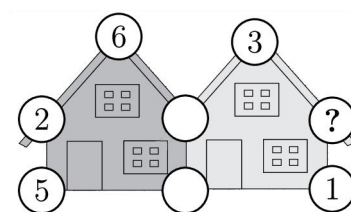
17. The sum of the five numbers in each house is 20. Some numbers have been painted over.

What number is hidden under the question mark?

měi jiān fáng zi shàng de wǔ gè shù zì zhī hé dōu shì 20, dàn yǒu xiē

shù zì bèi gài zhù le.

qǐng wèn wèn hào chù de shù zì shì jǐ

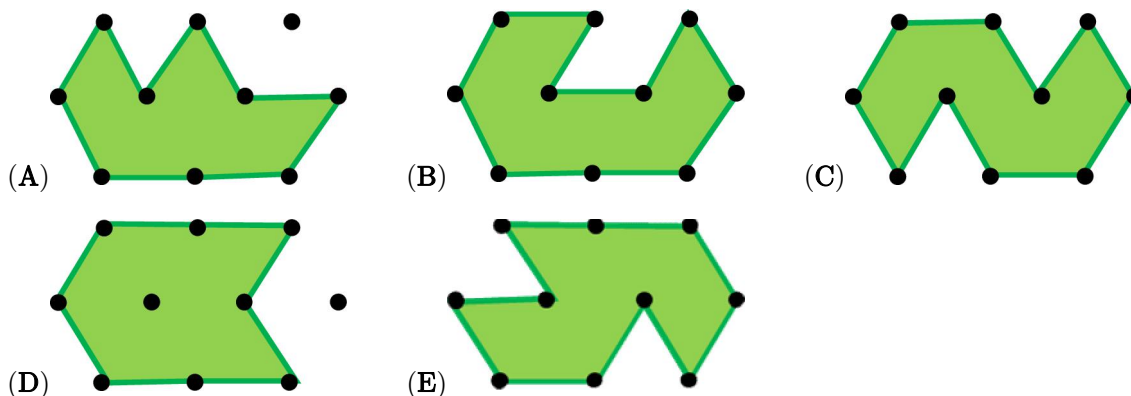


- (A) 3 (B) 4 (C) 7 (D) 9 (E) 14

18. Some lawns are shown below. Which lawn is the smallest?

xià miàn nǎ kuài cǎo dì zuì xiǎo

下面哪块草地最小?



19. Each year, Maria received teddy bears for her birthday. For her first birthday she received 1 teddy bear. For her second birthday she received 2 teddy bears. For each subsequent birthday she received one teddy bear more than the previous year.

How many teddy bears does Maria have in total when she is 6 years old?

měi nián xiǎo měi dōu huì shōu dào tài dí xióng zuò wéi shēng rì lǐ wù tā guò dì yī gè shēng rì

shí shōu dào le 1 zhī tài dí xióng guò dì èr gè shēng rì shí shōu dào le 2 zhī tài dí xióng zhī

hòu de měi gè shēng rì tā dōu bǐ shàng yī nián duō shōu dào yī zhī tài dí xióng

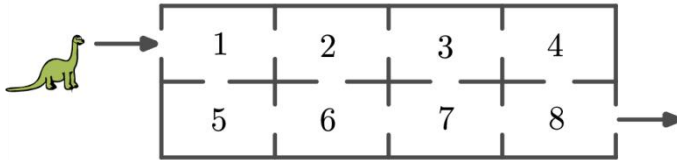
dāng xiǎo měi suì de shí hòu tā zǒng gòng yǒu duō shǎo zhī tài dí xióng

- (A) 19 (B) 20 (C) 21 (D) 22 (E) 23

20. Dino moves from the entrance to the exit by going through rooms. He can only go through each room once.

xiǎo kǒng lóng yào cóng rù kǒu jìn qù chuān guò yī xiē fáng jiān dào dá chū kǒu měi gè fáng jiān

zhǐ néng jīng guò yī cì



Dino adds up the numbers as he passes through each room. What is the highest total Dino can make?

xiǎo kǒng lóng zài jīng guò měi gè fáng jiān shí huì bǎ gāi fáng jiān de shù zì jiā qǐ lái qǐng wèn

tā néng dé dào de zuì dà shù zì zhī hé shì duō shǎo

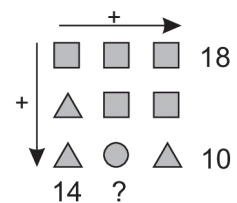
- (A) 27 (B) 29 (C) 32 (D) 34 (E) 36

21. In the picture, each shape stands for a different number.

Which number should be written in place of the question mark?

zài yòu tú zhōng měi zhǒng xíng zhuàng dài biǎo yī gè bù tóng de shù zì

qǐng wèn wèn hào chù yīng gāi tián nǎ ge shù zì



- (A) 10 (B) 12 (C) 14 (D) 16 (E) 18

22. Three zebras take part in a contest. The winner is the zebra with the most stripes. Runa has 15 stripes, Zara has 3 more than Runa. Runa has 5 fewer stripes than Biba. How many stripes does the winner have?

sān zhī bān mǎ cān jiā bǐ sài huò shèng zhě shì bān wén zuì duō de bān mǎ xiǎo yǒu tiáo bān

wén xiǎo bǐ xiǎo duō tiáo xiǎo bǐ xiǎo shǎo tiáo qǐng wèn huò shèng de bān mǎ yǒu

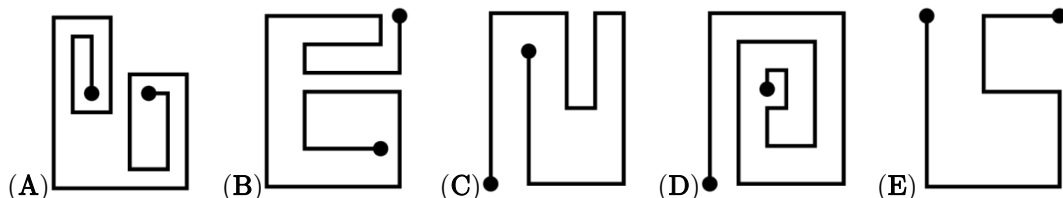
duō shǎo tiáo bān wén

- (A) 16 (B) 18 (C) 20 (D) 21 (E) 22

23. Kangy' s car can only turn left. It can never turn right. Which of the following five routes can Kangy take?

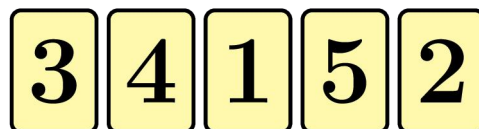
xiǎo kǎi de chē zhǐ néng zuǒ zhuǎn , bù néng yòu zhuǎn 。 yǐ xià wǔ tiáo lù xiàn zhōng , xiǎo kǎi kě yǐ

zǒu nǎ tiáo
走哪条?



24. There are five numbered cards on the table as shown.

You may swap two cards at each step. What is the smallest number of steps needed to put the cards into increasing order?



rú tú suǒ shì zhuō shàng yǒu wǔ zhāng biān hào de kǎ piàn
如 图 所 示， 桌 上 有 五 张 编 号 的 卡 片。

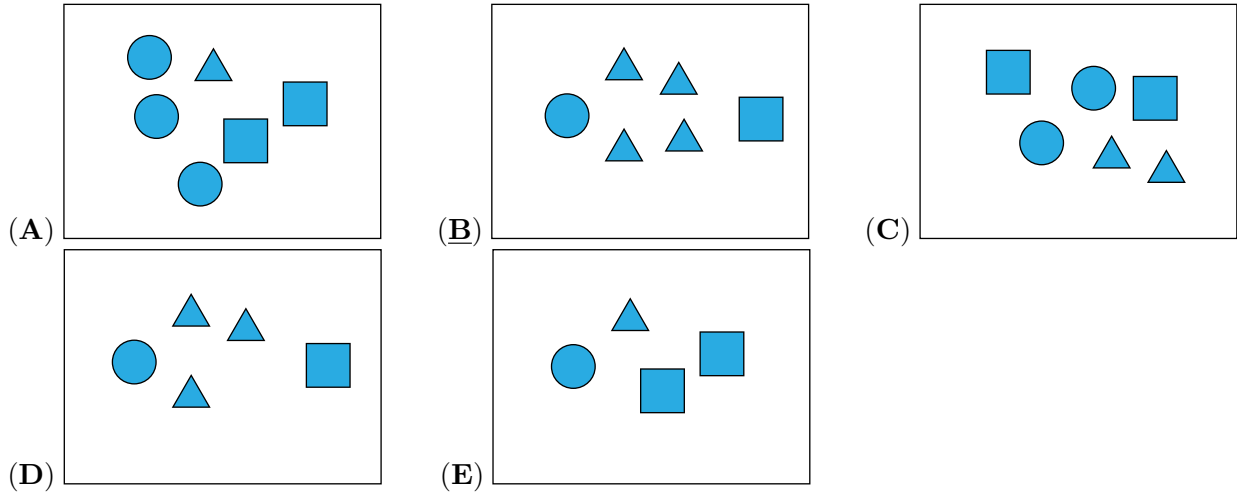
měi cì nǐ kě yǐ jiāo huàn liǎng zhāng kǎ piàn de wèi zhì yào jiāng kǎ piàn àn zhào cóng xiǎo dào

dà de shùn xù pái liè, zuì shǎo xū yào jiāo huàn jǐ cì?
大的顺序排列, 最少需要交换几次?

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

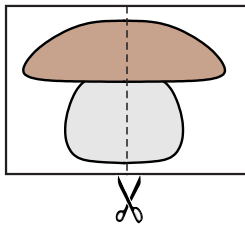
3 points

1 (FI). Which box contains the most triangles?

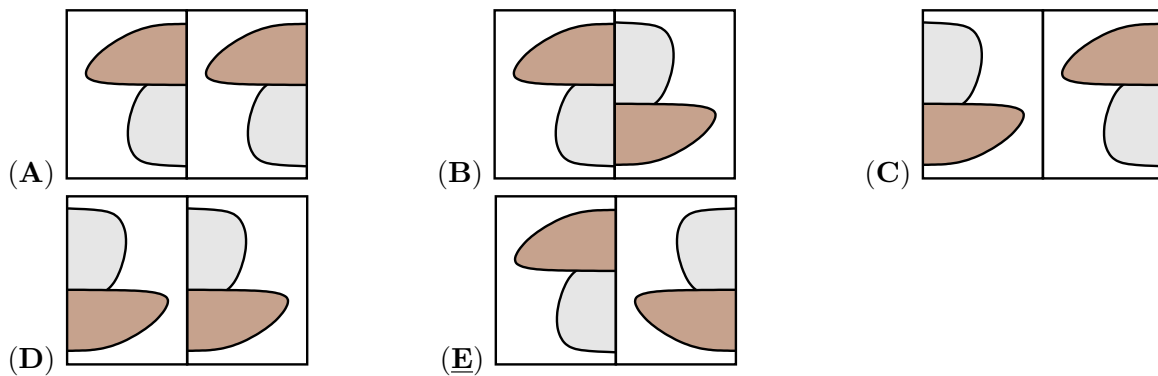


SOLUTION: Count the triangles in each option and see that option B has the most triangles.

2 (PL). Arek cuts this picture in half and puts the two pieces together.



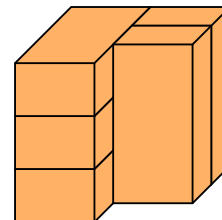
Which option shows the two pieces of Arek's picture?



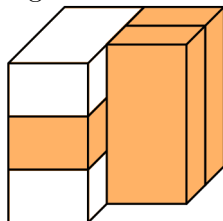
SOLUTION: Only option E contains the two pieces of the picture. You have to turn the right half of the picture E upside down.

3 (GR). The picture shows 5 identical bricks. How many bricks are touching exactly 3 other bricks?

- (A) 1 (B) 2 (C) 3
(D) 4 (E) 5

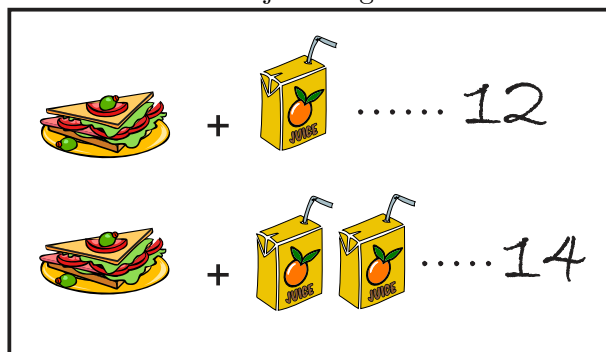


SOLUTION: By counting the bricks that touch exactly 3 others, we see that there are 2 - the white



bricks in the picture.

4 (BR). One sandwich and one juice together cost 12 euro. One sandwich and two juices together



cost 14 euro.

How much does one juice cost?

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

SOLUTION: Buying one more juice increases the price by 2. So each juice costs 2.

5 (DK). There has to be 2 coins in each row and each column.

			B
A		C	
		D	
E			

Where do you need to put the final coin?

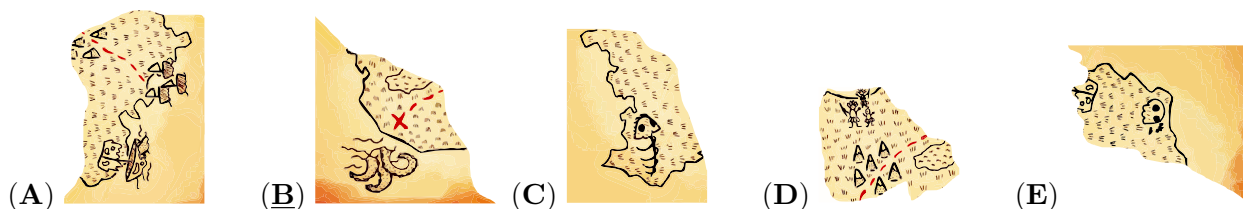
- (A) A (B) B (C) C (D) D (E) E

SOLUTION: Count the coins in every column and find the column with only one coin (third column). Second step: check letters C, D which fit another condition.

6 (UZ). A monkey has torn a piece from Captain Jack's map.

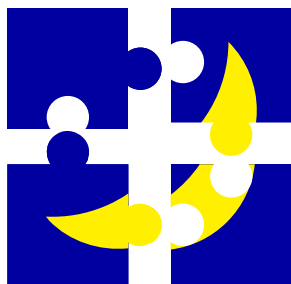
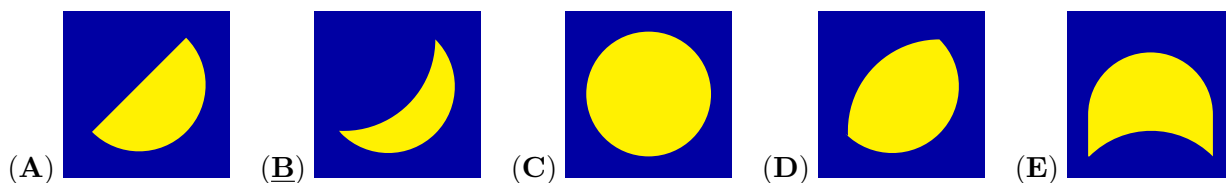
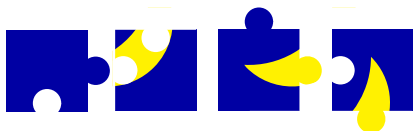


Which is the missing piece?



SOLUTION:

7 (DK). Peter puts the 4 puzzle pieces shown together to make a square. Which picture can he make?



SOLUTION:

8 (GR). Some ink spilled on a piece of squared paper, as shown in the picture.

How many of the squares have ink on them?

- (A) 16 (B) 17
(C) 18 (D) 19
(E) 20



SOLUTION: Counting one by one the answer is 20.

4 points

9 (GR). Kanga wrote down a number and then covered each digit with a shape. Different digits were covered by different shapes, and the same digits were covered by the same shape. Which number could be written under these shapes?



- (A) 34426 (B) 34526 (C) 34423 (D) 34424 (E) 32446

SOLUTION: Not 34526 because there are no repeating symbols. Not 34423 because first and last digit are the same. Not 34424 because the repeating digits are the same as the last digit. Not 32446 because the repeating digits are in the wrong position.

Solution 2.

We are looking for a number whose second and third digits are the same and the rest of the digits are different. This number is 34426.

10 (SK). One animal sleeps in each of the baskets. The koala and the fox are sleeping in baskets with the same pattern and shape. Which basket is the puppy sleeping in?

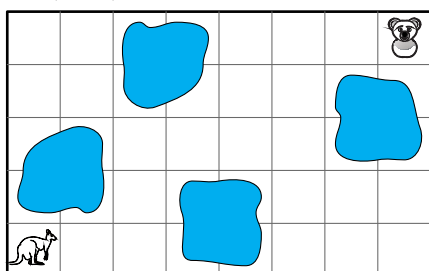


The kangaroo and the ostrich have the same pattern on their baskets. Which basket is the puppy sleeping in?

- (A) basket 1 (B) basket 2 (C) basket 3 (D) basket 4 (E) basket 5

SOLUTION: Koala and Fox sleep in baskets 2 and 4. Kangaroo and Ostrich sleep in 1 and 3. So puppy sleeps in basket 5.

11 (DK). Kanga wants to reach the koala without going through any of the coloured squares.



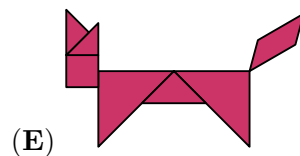
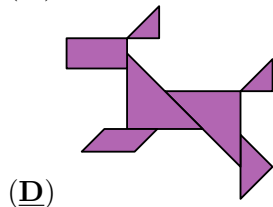
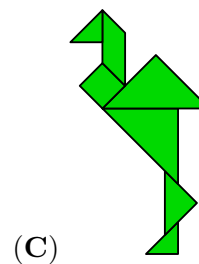
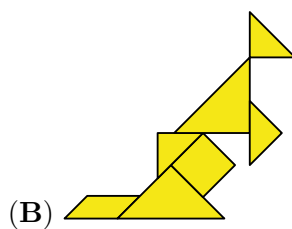
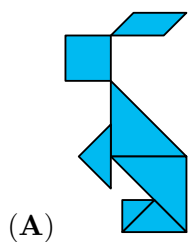
Which route could she take?

- (A) → → ↑ ↑ → → ↑ ↑ → → →
- (B) → → ↑ ↑ → → → → → ↑ ↑
- (C) → → ↑ ↑ ↑ ↑ → → → → →
- (D) → → ↑ ↑ → → ↑ ↑ ← ← ←
- (E) → → ↑ ↑ ↑ → → ↑ → → →

SOLUTION: The kangaroo can only start by jumping right, right, which is the same for all options. Then the kangaroo must jump up, up, right, which rules out options C and E. The last jump must be to the right, so B and D are ruled out too. We finish by checking that option A works.

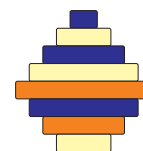
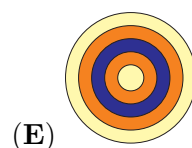
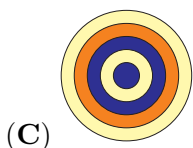
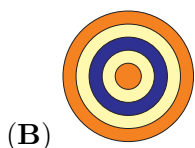
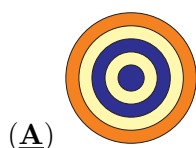
12 (IR). In one of the pictures below, a shape is used that cannot be seen in the others. In which

picture is it?



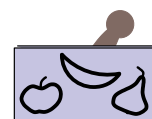
SOLUTION: The rectangle appears only in option D.

13 (KW). Which option shows the view from above this stack of discs?



SOLUTION: In the top view we can only see the top 5 discs. The the order of the colours from top to bottom is: blue, yellow, blue, green, red. The answer is A.

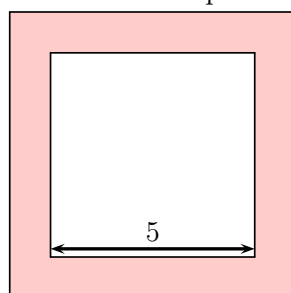
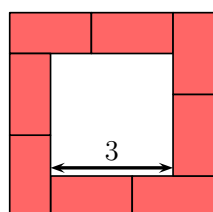
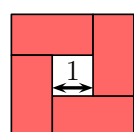
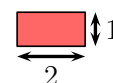
14 (PL). Which of the following pictures will we see when we use the stamp shown?



SOLUTION: The answer is D, because it's like a mirror.

So if in the original we have apple, banana and pear, in the stamp we will have the opposite order (pear, banana, and apple). The petioles of the pear and apple should face each other (should point towards the banana).

15 (DK). Katrin builds a path around each square using tiles like the one shown.



How many tiles does she use around a square with side 5?

(A) 10

(B) 11

(C) 12

(D) 14

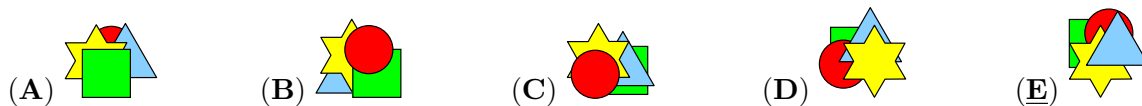
(E) 16

SOLUTION: The path needs one more tile along each edge and so 4 more tiles than the square of side length 5. The answer is 12.

- 16 (PL). Ann has 4 stickers as shown. 

She sticks down the star after she sticks down the square. She sticks down the star before she sticks down the triangle.

Which picture could she end up with?

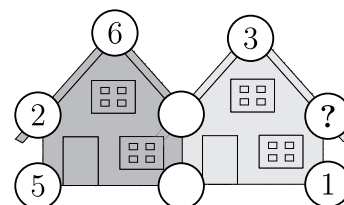


SOLUTION: Ann sticks down the star after she sticks down the square. And she sticks down the star before the triangle. So the square is under the star and the triangle is on top of the star. Such situation is shown only in option E.

5 points

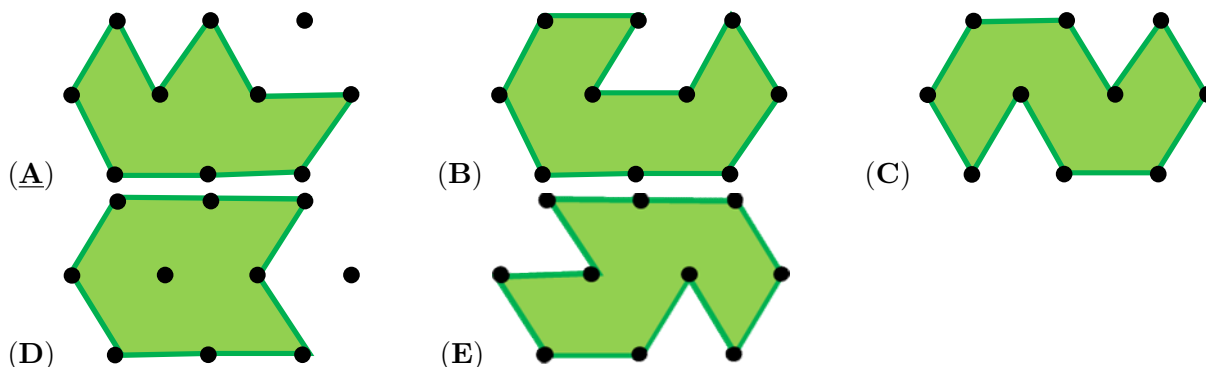
- 17 (PL). The sum of the five numbers in each house is 20. Some numbers have been painted over. What number is hidden under the question mark?

- (A) 3 (B) 4 (C) 7
(D) 9 (E) 14



SOLUTION: The sum of the visible numbers in the first house is $6 + 2 + 5 = 13$. The sum of the numbers in the other two circles in this house is equal to $20 - 13 = 7$. The sum of the visible numbers in the second house is $3 + 1 = 4$. So, in the circle with the question mark there is $20 - 4 - 7 = 9$. Another solution: The information that the sum of the numbers in each house is equal to 20 is not needed. It is enough to note that the sums of the remaining numbers in these houses are equal. Therefore $6 + 2 + 5 = 3 + 1 + ?$, and so $? = 9$.

- 18 (GR). Some lawns are shown below. Which lawn is the smallest?



SOLUTION: If you connect the points with lines, you can build triangles of equal size. Option A only has 7 triangles, so the lawn is smaller. The other options have 8 triangles.

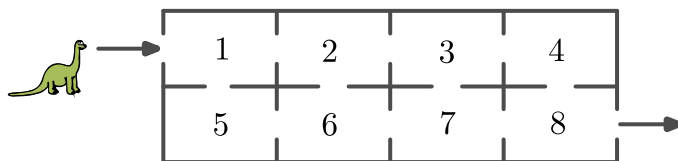
- 19 (MK). Each year, Maria received teddy bears for her birthday. For her first birthday she received 1 teddy bear. For her second birthday she received 2 teddy bears. For each subsequent birthday she received one teddy bear more than the previous year.

How many teddy bears does Maria have in total when she is 6 years old?

- (A) 19 (B) 20 (C) 21 (D) 22 (E) 23

SOLUTION: The number of teddies is $1 + 2 + 3 + 4 + 5 + 6 = 21$.

20 (CT). Dino moves from the entrance to the exit by going through rooms. He can only go through each room once.



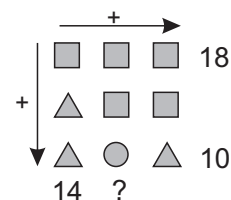
Dino adds up the numbers as he passes through each room. What is the highest total Dino can make?

- (A) 27 (B) 29 (C) 32 (D) 34 (E) 36

SOLUTION: Dino goes through rooms 1 - 5 - 6 - 7 - 3 - 4 - 8. This gives him the highest total, which is 34.

21 (EE). In the picture, each shape stands for a different number. Which number should be written in place of the question mark?

- (A) 10 (B) 12 (C) 14
(D) 16 (E) 18



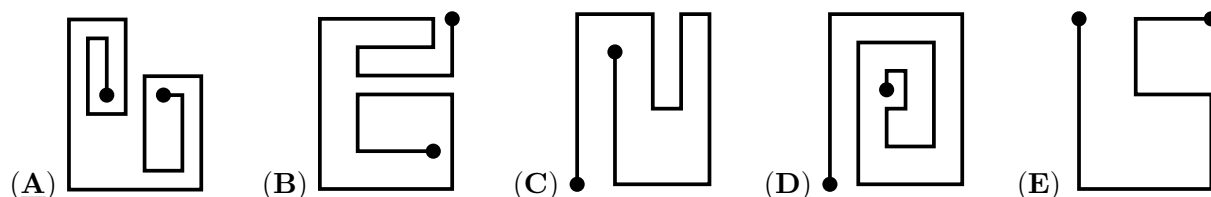
SOLUTION: Three squares together make 18, so each square is 6. One square and two triangles make 14, so each triangle is 4. Two triangles and one circle make 10, so each circle is 2. Then two squares and one circle make 14.

22 (SK). Three zebras take part in a contest. The winner is the zebra with the most stripes. Runa has 15 stripes, Zara has 3 more than Runa. Runa has 5 fewer stripes than Biba. How many stripes does the winner have?

- (A) 16 (B) 18 (C) 20 (D) 21 (E) 22

SOLUTION: Runa has 15 stripes, Zara has 18 stripes and Biba has 20 stripes.

23 (HK). Kangy's car can only turn left. It can never turn right. Which of the following five routes can Kangy take?



SOLUTION: All options, except for A, have a right turn.

24 (HU). There are five numbered cards on the table as shown.

You may swap two cards at each step. What is the smallest number of steps needed to put the cards into increasing order?



- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5

SOLUTION: None of the cards are in their final place. Only two cards can be put in their final place in one step, so a minimum of 3 steps are required. 3 steps are enough, for example: swap 3 and 1, swap 4 and 2, swap 5 and 4.